

# 10th Grade Midterm Analysis of Decisions: C4–C5

## Solutions

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### Evaluated criteria

- C4** Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.
- C5** Constructs opportunity loss tables and applies the Minimax regret criterion to select the alternative that minimises the maximum oport

*All payoffs are in thousands of dollars.*  
*Regret = best payoff in the state – payoff.*

## 1 Introduction

This section presents the general procedure for analysing a payoff table under uncertainty. The alternatives are denoted by  $A_i$ , the states of nature by  $S_j$ , and the payoff obtained by choosing alternative  $A_i$  when state  $S_j$  occurs by

$$P(A_i, S_j).$$

The analysis uses three decision criteria:

1. **Maximax:** choose the alternative with the largest possible payoff.
2. **Maximin:** choose the alternative with the largest worst-case payoff.
3. **Minimax Regret:** choose the alternative whose maximum opportunity loss is the smallest.

Because the payoffs in this section are represented symbolically, the final choice depends on the numerical values assigned to the expressions  $P(A_i, S_j)$ .

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### Problems:

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#### 1.1. General $2 \times 2$ Problem

#### 1.2. General $4 \times 3$ Problem

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### 1.1 General $2 \times 2$ Problem

A decision maker must choose between two alternatives,  $A_1$  and  $A_2$ . Two states of nature,  $S_1$  and  $S_2$ , may occur. The payoff associated with each alternative–state combination is represented by  $P(A_i, S_j)$ , where  $i$  identifies the alternative and  $j$  identifies the state of nature.

| <i>General payoff table</i> |                   |                   |
|-----------------------------|-------------------|-------------------|
| State of nature             | Alternative $A_1$ | Alternative $A_2$ |
| State $S_1$                 | $P(A_1, S_1)$     | $P(A_2, S_1)$     |
| State $S_2$                 | $P(A_1, S_2)$     | $P(A_2, S_2)$     |

Use Maximax, Maximin, and Minimax Regret to determine the decision rules for this general payoff table.

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

#### Step 1: Find the largest payoff for each alternative.

For  $A_1$ , compare its payoffs under  $S_1$  and  $S_2$ :

$$M_1 = \max\{P(A_1, S_1), P(A_1, S_2)\}.$$

For  $A_2$ , compare its payoffs under  $S_1$  and  $S_2$ :

$$M_2 = \max\{P(A_2, S_1), P(A_2, S_2)\}.$$

Therefore,

$$M_1 = \max\{P(A_1, S_1), P(A_1, S_2)\},$$

$$M_2 = \max\{P(A_2, S_1), P(A_2, S_2)\}.$$

**Step 2: Apply the Maximax criterion.**

Compare the largest payoffs  $M_1$  and  $M_2$ :

$$M^* = \max\{M_1, M_2\}.$$

Equivalently,

$$M^* = \max\{\max\{P(A_1, S_1), P(A_1, S_2)\}, \max\{P(A_2, S_1), P(A_2, S_2)\}\}.$$

The Maximax decision is

$$A_{\text{Maximax}} = \arg \max_{i \in \{1,2\}} \left\{ \max_{j \in \{1,2\}} P(A_i, S_j) \right\}.$$

Thus, choose  $A_1$  if  $M_1 > M_2$ , choose  $A_2$  if  $M_2 > M_1$ , and report a tie if  $M_1 = M_2$ .

**Step 3: Find the smallest payoff for each alternative.**

For  $A_1$ ,

$$m_1 = \min\{P(A_1, S_1), P(A_1, S_2)\}.$$

For  $A_2$ ,

$$m_2 = \min\{P(A_2, S_1), P(A_2, S_2)\}.$$

Therefore,

$$m_1 = \min\{P(A_1, S_1), P(A_1, S_2)\},$$

$$m_2 = \min\{P(A_2, S_1), P(A_2, S_2)\}.$$

**Step 4: Apply the Maximin criterion.**

Compare the smallest payoffs  $m_1$  and  $m_2$ :

$$m^* = \max\{m_1, m_2\}.$$

Equivalently,

$$m^* = \max\{\min\{P(A_1, S_1), P(A_1, S_2)\}, \min\{P(A_2, S_1), P(A_2, S_2)\}\}.$$

The Maximin decision is

$$A_{\text{Maximin}} = \arg \max_{i \in \{1,2\}} \left\{ \min_{j \in \{1,2\}} P(A_i, S_j) \right\}.$$

Thus, choose  $A_1$  if  $m_1 > m_2$ , choose  $A_2$  if  $m_2 > m_1$ , and report a tie if  $m_1 = m_2$ .

**C5 - Constructs opportunity loss tables and applies the Minimax regret criterion**

**Step 1: Find the best payoff in each state of nature.**

Under  $S_1$ , compare the payoffs of  $A_1$  and  $A_2$ :

$$b_1 = \max\{P(A_1, S_1), P(A_2, S_1)\}.$$

Under  $S_2$ , compare the payoffs of  $A_1$  and  $A_2$ :

$$b_2 = \max\{P(A_1, S_2), P(A_2, S_2)\}.$$

Therefore,

$$b_1 = \max\{P(A_1, S_1), P(A_2, S_1)\},$$

$$b_2 = \max\{P(A_1, S_2), P(A_2, S_2)\}.$$

**Step 2: Calculate the regret for every alternative–state combination.**

The regret associated with alternative  $A_i$  under state  $S_j$  is

$$r_{ij} = b_j - P(A_i, S_j).$$

Consequently,

$$r_{11} = b_1 - P(A_1, S_1), \quad r_{12} = b_2 - P(A_1, S_2),$$

$$r_{21} = b_1 - P(A_2, S_1), \quad r_{22} = b_2 - P(A_2, S_2).$$

The general regret table is therefore

| Alternative | State $S_1$         | State $S_2$         |
|-------------|---------------------|---------------------|
| $A_1$       | $b_1 - P(A_1, S_1)$ | $b_2 - P(A_1, S_2)$ |
| $A_2$       | $b_1 - P(A_2, S_1)$ | $b_2 - P(A_2, S_2)$ |

**Step 3: Find the maximum regret for each alternative.**

For  $A_1$ ,

$$R_1 = \max\{b_1 - P(A_1, S_1), b_2 - P(A_1, S_2)\}.$$

For  $A_2$ ,

$$R_2 = \max\{b_1 - P(A_2, S_1), b_2 - P(A_2, S_2)\}.$$

Thus,

$$R_1 = \max\{r_{11}, r_{12}\},$$

$$R_2 = \max\{r_{21}, r_{22}\}.$$

**Step 4: Apply the Minimax Regret criterion.**

Compare the maximum regrets  $R_1$  and  $R_2$ :

$$R^* = \min\{R_1, R_2\}.$$

The Minimax Regret decision is

$$A_{\text{Minimax Regret}} = \arg \min_{i \in \{1,2\}} \left\{ \max_{j \in \{1,2\}} [b_j - P(A_i, S_j)] \right\}.$$

Thus, choose  $A_1$  if  $R_1 < R_2$ , choose  $A_2$  if  $R_2 < R_1$ , and report a tie if  $R_1 = R_2$ .

## General decisions summary

| Criterion      | General decision rule                                                |
|----------------|----------------------------------------------------------------------|
| Maximax        | $\arg \max_{i \in \{1,2\}} \max_{j \in \{1,2\}} P(A_i, S_j)$         |
| Maximin        | $\arg \max_{i \in \{1,2\}} \min_{j \in \{1,2\}} P(A_i, S_j)$         |
| Minimax Regret | $\arg \min_{i \in \{1,2\}} \max_{j \in \{1,2\}} [b_j - P(A_i, S_j)]$ |



## 1.2 General $4 \times 3$ Problem

A decision maker must choose among three alternatives,  $A_1$ ,  $A_2$ , and  $A_3$ . Four states of nature,  $S_1$ ,  $S_2$ ,  $S_3$ , and  $S_4$ , may occur. The payoff associated with each alternative–state combination is represented by  $P(A_i, S_j)$ .

| <i>General payoff table</i> |                   |                   |                   |
|-----------------------------|-------------------|-------------------|-------------------|
| State of nature             | Alternative $A_1$ | Alternative $A_2$ | Alternative $A_3$ |
| State $S_1$                 | $P(A_1, S_1)$     | $P(A_2, S_1)$     | $P(A_3, S_1)$     |
| State $S_2$                 | $P(A_1, S_2)$     | $P(A_2, S_2)$     | $P(A_3, S_2)$     |
| State $S_3$                 | $P(A_1, S_3)$     | $P(A_2, S_3)$     | $P(A_3, S_3)$     |
| State $S_4$                 | $P(A_1, S_4)$     | $P(A_2, S_4)$     | $P(A_3, S_4)$     |

Use Maximax, Maximin, and Minimax Regret to determine the decision rules for this general payoff table.

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

#### Step 1: Find the largest payoff for each alternative.

For the three alternatives,

$$M_1 = \max\{P(A_1, S_1), P(A_1, S_2), P(A_1, S_3), P(A_1, S_4)\},$$

$$M_2 = \max\{P(A_2, S_1), P(A_2, S_2), P(A_2, S_3), P(A_2, S_4)\},$$

$$M_3 = \max\{P(A_3, S_1), P(A_3, S_2), P(A_3, S_3), P(A_3, S_4)\}.$$

#### Step 2: Apply the Maximax criterion.

Compare the three largest payoffs:

$$M^* = \max\{M_1, M_2, M_3\}.$$

Equivalently,

$$M^* = \max\left\{\max\{P(A_1, S_1), P(A_1, S_2), P(A_1, S_3), P(A_1, S_4)\},\right. \\ \left.\max\{P(A_2, S_1), P(A_2, S_2), P(A_2, S_3), P(A_2, S_4)\},\right. \\ \left.\max\{P(A_3, S_1), P(A_3, S_2), P(A_3, S_3), P(A_3, S_4)\}\right\}.$$

The Maximax decision is

$$A_{\text{Maximax}} = \arg \max_{i \in \{1,2,3\}} \left\{ \max_{j \in \{1,2,3,4\}} P(A_i, S_j) \right\}.$$

Choose the alternative whose value  $M_i$  is the largest. If two or more alternatives have the same largest value, the Maximax criterion produces a tie.

**Step 3: Find the smallest payoff for each alternative.**

For the three alternatives,

$$m_1 = \min\{P(A_1, S_1), P(A_1, S_2), P(A_1, S_3), P(A_1, S_4)\}, \\ m_2 = \min\{P(A_2, S_1), P(A_2, S_2), P(A_2, S_3), P(A_2, S_4)\}, \\ m_3 = \min\{P(A_3, S_1), P(A_3, S_2), P(A_3, S_3), P(A_3, S_4)\}.$$

**Step 4: Apply the Maximin criterion.**

Compare the three smallest payoffs:

$$m^* = \max\{m_1, m_2, m_3\}.$$

Equivalently,

$$m^* = \max\left\{\min\{P(A_1, S_1), P(A_1, S_2), P(A_1, S_3), P(A_1, S_4)\},\right. \\ \left.\min\{P(A_2, S_1), P(A_2, S_2), P(A_2, S_3), P(A_2, S_4)\},\right. \\ \left.\min\{P(A_3, S_1), P(A_3, S_2), P(A_3, S_3), P(A_3, S_4)\}\right\}.$$

The Maximin decision is

$$A_{\text{Maximin}} = \arg \max_{i \in \{1,2,3\}} \left\{ \min_{j \in \{1,2,3,4\}} P(A_i, S_j) \right\}.$$

Choose the alternative whose value  $m_i$  is the largest. If two or more alternatives have the same largest value, the Maximin criterion produces a tie.

**C5 - Constructs opportunity loss tables and applies the Minimax regret criterion**

**Step 1: Find the best payoff in each state of nature.**

For the four states,

$$b_1 = \max\{P(A_1, S_1), P(A_2, S_1), P(A_3, S_1)\}, \\ b_2 = \max\{P(A_1, S_2), P(A_2, S_2), P(A_3, S_2)\}, \\ b_3 = \max\{P(A_1, S_3), P(A_2, S_3), P(A_3, S_3)\}, \\ b_4 = \max\{P(A_1, S_4), P(A_2, S_4), P(A_3, S_4)\}.$$

**Step 2: Calculate the regret for every alternative–state combination.**

For every alternative  $A_i$  and state  $S_j$ , the regret is

$$r_{ij} = b_j - P(A_i, S_j).$$

The general regret table is therefore

| Alternative | State $S_1$         | State $S_2$         | State $S_3$         | State $S_4$         |
|-------------|---------------------|---------------------|---------------------|---------------------|
| $A_1$       | $b_1 - P(A_1, S_1)$ | $b_2 - P(A_1, S_2)$ | $b_3 - P(A_1, S_3)$ | $b_4 - P(A_1, S_4)$ |
| $A_2$       | $b_1 - P(A_2, S_1)$ | $b_2 - P(A_2, S_2)$ | $b_3 - P(A_2, S_3)$ | $b_4 - P(A_2, S_4)$ |
| $A_3$       | $b_1 - P(A_3, S_1)$ | $b_2 - P(A_3, S_2)$ | $b_3 - P(A_3, S_3)$ | $b_4 - P(A_3, S_4)$ |

**Step 3: Find the maximum regret for each alternative.**

For  $A_1$ ,

$$R_1 = \max\{b_1 - P(A_1, S_1), b_2 - P(A_1, S_2), b_3 - P(A_1, S_3), b_4 - P(A_1, S_4)\}.$$

For  $A_2$ ,

$$R_2 = \max\{b_1 - P(A_2, S_1), b_2 - P(A_2, S_2), b_3 - P(A_2, S_3), b_4 - P(A_2, S_4)\}.$$

For  $A_3$ ,

$$R_3 = \max\{b_1 - P(A_3, S_1), b_2 - P(A_3, S_2), b_3 - P(A_3, S_3), b_4 - P(A_3, S_4)\}.$$

Equivalently,

$$R_1 = \max\{r_{11}, r_{12}, r_{13}, r_{14}\},$$

$$R_2 = \max\{r_{21}, r_{22}, r_{23}, r_{24}\},$$

$$R_3 = \max\{r_{31}, r_{32}, r_{33}, r_{34}\}.$$

**Step 4: Apply the Minimax Regret criterion.**

Compare the maximum regrets:

$$R^* = \min\{R_1, R_2, R_3\}.$$

The Minimax Regret decision is

$$A_{\text{Minimax Regret}} = \arg \min_{i \in \{1,2,3\}} \left\{ \max_{j \in \{1,2,3,4\}} [b_j - P(A_i, S_j)] \right\}.$$

Choose the alternative whose value  $R_i$  is the smallest. If two or more alternatives have the same smallest value, the Minimax Regret criterion produces a tie.

**General decisions summary**

| Criterion      | General decision rule                                                      |
|----------------|----------------------------------------------------------------------------|
| Maximax        | $\arg \max_{i \in \{1,2,3\}} \max_{j \in \{1,2,3,4\}} P(A_i, S_j)$         |
| Maximin        | $\arg \max_{i \in \{1,2,3\}} \min_{j \in \{1,2,3,4\}} P(A_i, S_j)$         |
| Minimax Regret | $\arg \min_{i \in \{1,2,3\}} \max_{j \in \{1,2,3,4\}} [b_j - P(A_i, S_j)]$ |



## 2 Problems $2 \times 2$

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### Problems:

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2.1. Food Truck Location (ABB)

2.2. Promotional Campaign (ABA)

2.3. Equipment Lease (AAA)

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### 2.1 Food Truck Location

A food truck operator must choose between Location A and Location B. Demand can be strong or weak. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a location.

| <i>Payoff table</i> |            |            |
|---------------------|------------|------------|
| State of nature     | Location A | Location B |
| Strong demand       | 100        | 60         |
| Weak demand         | 0          | 50         |

Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

**C4 - Applies the Maximin and Maximax criteria**

$$\max(\text{Location A}) = \max\{100, 0\} = 100,$$

$$\max(\text{Location B}) = \max\{60, 50\} = 60.$$

$$\max\{\max(\text{Location A}), \max(\text{Location B})\} = \max\{100, 60\} = 100.$$

The Maximax choice is Location A because it has the highest possible payoff.

$$\min(\text{Location A}) = \min\{100, 0\} = 0,$$

$$\min(\text{Location B}) = \min\{60, 50\} = 50.$$

$$\max\{\min(\text{Location A}), \min(\text{Location B})\} = \max\{0, 50\} = 50.$$

The Maximin choice is Location B because it has the largest worst-case payoff.

**C5 - Constructs opportunity loss tables and applies the Minimax regret criterion**

Best payoff in each state:

$$\text{Strong demand: } \max\{100, 60\} = 100,$$

$$\text{Weak demand: } \max\{0, 50\} = 50.$$

Regret table (best payoff – payoff):

| Alternative | Strong demand   | Weak demand   | Maximum regret |
|-------------|-----------------|---------------|----------------|
| Location A  | $100 - 100 = 0$ | $50 - 0 = 50$ | 50             |
| Location B  | $100 - 60 = 40$ | $50 - 50 = 0$ | 40             |

The Minimax Regret choice is Location B because it has the smallest maximum regret, 40.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Location A     |
| Maximin        | Location B     |
| Minimax Regret | Location B     |



## 2.2 Promotional Campaign

A business must choose between Campaign A and Campaign B. Customer response can be strong or weak. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a campaign.

| <i>Payoff table</i> |            |            |
|---------------------|------------|------------|
| State of nature     | Campaign A | Campaign B |
| Strong response     | 100        | 80         |
| Weak response       | 20         | 30         |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Campaign A}) = \max\{100, 20\} = 100,$$

$$\max(\text{Campaign B}) = \max\{80, 30\} = 80.$$

$$\max\{\max(\text{Campaign A}), \max(\text{Campaign B})\} = \max\{100, 80\} = 100.$$

The Maximax choice is Campaign A because it has the highest possible payoff.

$$\min(\text{Campaign A}) = \min\{100, 20\} = 20,$$

$$\min(\text{Campaign B}) = \min\{80, 30\} = 30.$$

$$\max\{\min(\text{Campaign A}), \min(\text{Campaign B})\} = \max\{20, 30\} = 30.$$

The Maximin choice is Campaign B because it has the largest worst-case payoff.

### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Strong response: } \max\{100, 80\} = 100,$$

$$\text{Weak response: } \max\{20, 30\} = 30.$$

Regret table (best payoff – payoff):

| Alternative | Strong response | Weak response  | Maximum regret |
|-------------|-----------------|----------------|----------------|
| Campaign A  | $100 - 100 = 0$ | $30 - 20 = 10$ | 10             |
| Campaign B  | $100 - 80 = 20$ | $30 - 30 = 0$  | 20             |

The Minimax Regret choice is Campaign A because it has the smallest maximum regret, 10.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Campaign A     |
| Maximin        | Campaign B     |
| Minimax Regret | Campaign A     |



## 2.3 Equipment Lease

A company must choose between Lease A and Lease B. Market conditions can be favourable or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a lease.

| <i>Payoff table</i> |         |         |
|---------------------|---------|---------|
| State of nature     | Lease A | Lease B |
| Favourable market   | 70      | 60      |
| Unfavourable market | 40      | 30      |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Lease A}) = \max\{70, 40\} = 70,$$

$$\max(\text{Lease B}) = \max\{60, 30\} = 60.$$

$$\max\{\max(\text{Lease A}), \max(\text{Lease B})\} = \max\{70, 60\} = 70.$$

The Maximax choice is Lease A because it has the highest possible payoff.

$$\min(\text{Lease A}) = \min\{70, 40\} = 40,$$

$$\min(\text{Lease B}) = \min\{60, 30\} = 30.$$

$$\max\{\min(\text{Lease A}), \min(\text{Lease B})\} = \max\{40, 30\} = 40.$$

The Maximin choice is Lease A because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Favourable market: } \max\{70, 60\} = 70,$$

$$\text{Unfavourable market: } \max\{40, 30\} = 40.$$

Regret table (best payoff – payoff):

| Alternative | Favourable     | Unfavourable   | Maximum regret |
|-------------|----------------|----------------|----------------|
| Lease A     | $70 - 70 = 0$  | $40 - 40 = 0$  | 0              |
| Lease B     | $70 - 60 = 10$ | $40 - 30 = 10$ | 10             |

The Minimax Regret choice is Lease A because it has the smallest maximum regret, 0.

#### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Lease A        |
| Maximin        | Lease A        |
| Minimax Regret | Lease A        |



### 3 Problems $2 \times 3$

#### Problems:

3.1. Production Plan (ABB)

3.2. Marketing Strategy (ABA)

3.3. Event Venue (ABC)

3.4. Distribution Plan (AAA)

### 3.1 Production Plan

A manufacturer must choose Production Plan A, Plan B, or Plan C. Demand can be strong or weak. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a plan.

| <i>Payoff table</i> |        |        |        |
|---------------------|--------|--------|--------|
| State of nature     | Plan A | Plan B | Plan C |
| Strong demand       | 100    | 70     | 80     |
| Weak demand         | 0      | 60     | 20     |

**Questions/Tasks.**

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

**C4 - Applies the Maximin and Maximax criteria**

$$\max(\text{Plan A}) = \max\{100, 0\} = 100,$$

$$\max(\text{Plan B}) = \max\{70, 60\} = 70,$$

$$\max(\text{Plan C}) = \max\{80, 20\} = 80.$$

$$\max\{\max(\text{Plan A}), \max(\text{Plan B}), \max(\text{Plan C})\} = \max\{100, 70, 80\} = 100.$$

The Maximax choice is Plan A because it has the highest possible payoff.

$$\min(\text{Plan A}) = \min\{100, 0\} = 0,$$

$$\min(\text{Plan B}) = \min\{70, 60\} = 60,$$

$$\min(\text{Plan C}) = \min\{80, 20\} = 20.$$

$$\max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C})\} = \max\{0, 60, 20\} = 60.$$

The Maximin choice is Plan B because it has the largest worst-case payoff.

**C5 - Constructs opportunity loss tables and applies the Minimax regret criterion**

Best payoff in each state:

$$\text{Strong demand: } \max\{100, 70, 80\} = 100,$$

$$\text{Weak demand: } \max\{0, 60, 20\} = 60.$$

Regret table (best payoff – payoff):

| Alternative | Strong demand   | Weak demand    | Maximum regret |
|-------------|-----------------|----------------|----------------|
| Plan A      | $100 - 100 = 0$ | $60 - 0 = 60$  | 60             |
| Plan B      | $100 - 70 = 30$ | $60 - 60 = 0$  | 30             |
| Plan C      | $100 - 80 = 20$ | $60 - 20 = 40$ | 40             |

The Minimax Regret choice is Plan B because it has the smallest maximum regret, 30.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Plan A         |
| Maximin        | Plan B         |
| Minimax Regret | Plan B         |



### 3.2 Marketing Strategy

A retailer must select Strategy A, Strategy B, or Strategy C for a new marketing campaign. Customer response can be strong or weak. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a strategy.

| <i>Payoff table</i> |            |            |            |
|---------------------|------------|------------|------------|
| State of nature     | Strategy A | Strategy B | Strategy C |
| Strong response     | 100        | 80         | 70         |
| Weak response       | 40         | 50         | 45         |

#### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Strategy A}) = \max\{100, 40\} = 100,$$

$$\max(\text{Strategy B}) = \max\{80, 50\} = 80,$$

$$\max(\text{Strategy C}) = \max\{70, 45\} = 70.$$

$$\max\{\max(\text{Strategy A}), \max(\text{Strategy B}), \max(\text{Strategy C})\} = \max\{100, 80, 70\} = 100.$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\min(\text{Strategy A}) = \min\{100, 40\} = 40,$$

$$\min(\text{Strategy B}) = \min\{80, 50\} = 50,$$

$$\min(\text{Strategy C}) = \min\{70, 45\} = 45.$$

$$\max\{\min(\text{Strategy A}), \min(\text{Strategy B}), \min(\text{Strategy C})\} = \max\{40, 50, 45\} = 50.$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

## C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

Strong response:  $\max\{100, 80, 70\} = 100$ ,

Weak response:  $\max\{40, 50, 45\} = 50$ .

Regret table (best payoff – payoff):

| Alternative | Strong response | Weak response  | Maximum regret |
|-------------|-----------------|----------------|----------------|
| Strategy A  | $100 - 100 = 0$ | $50 - 40 = 10$ | 10             |
| Strategy B  | $100 - 80 = 20$ | $50 - 50 = 0$  | 20             |
| Strategy C  | $100 - 70 = 30$ | $50 - 45 = 5$  | 30             |

The Minimax Regret choice is Strategy A because it has the smallest maximum regret, 10.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Strategy A     |
| Maximin        | Strategy B     |
| Minimax Regret | Strategy A     |



## 3.3 Event Venue

An event organiser must select Venue A, Venue B, or Venue C. Attendance can be high or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a venue.

| <i>Payoff table</i> |         |         |         |
|---------------------|---------|---------|---------|
| State of nature     | Venue A | Venue B | Venue C |
| High attendance     | 100     | 70      | 80      |
| Low attendance      | 0       | 60      | 50      |

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Venue A}) = \max\{100, 0\} = 100,$$

$$\max(\text{Venue B}) = \max\{70, 60\} = 70,$$

$$\max(\text{Venue C}) = \max\{80, 50\} = 80.$$

$$\max\{\max(\text{Venue A}), \max(\text{Venue B}), \max(\text{Venue C})\} = \max\{100, 70, 80\} = 100.$$

The Maximax choice is Venue A because it has the highest possible payoff.

$$\min(\text{Venue A}) = \min\{100, 0\} = 0,$$

$$\min(\text{Venue B}) = \min\{70, 60\} = 60,$$

$$\min(\text{Venue C}) = \min\{80, 50\} = 50.$$

$$\max\{\min(\text{Venue A}), \min(\text{Venue B}), \min(\text{Venue C})\} = \max\{0, 60, 50\} = 60.$$

The Maximin choice is Venue B because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High attendance: } \max\{100, 70, 80\} = 100,$$

$$\text{Low attendance: } \max\{0, 60, 50\} = 60.$$

Regret table (best payoff – payoff):

| Alternative | High attendance | Low attendance | Maximum regret |
|-------------|-----------------|----------------|----------------|
| Venue A     | $100 - 100 = 0$ | $60 - 0 = 60$  | 60             |
| Venue B     | $100 - 70 = 30$ | $60 - 60 = 0$  | 30             |
| Venue C     | $100 - 80 = 20$ | $60 - 50 = 10$ | 20             |

The Minimax Regret choice is Venue C because it has the smallest maximum regret, 20.

#### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Venue A        |
| Maximin        | Venue B        |
| Minimax Regret | Venue C        |



### 3.4 Distribution Plan

A company must select Distribution Plan A, Plan B, or Plan C. Market conditions can be favourable or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a distribution plan.

| <i>Payoff table</i> |        |        |        |
|---------------------|--------|--------|--------|
| State of nature     | Plan A | Plan B | Plan C |
| Favourable market   | 90     | 80     | 70     |
| Unfavourable market | 60     | 40     | 50     |

#### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Plan A}) = \max\{90, 60\} = 90,$$

$$\max(\text{Plan B}) = \max\{80, 40\} = 80,$$

$$\max(\text{Plan C}) = \max\{70, 50\} = 70.$$

$$\max\{\max(\text{Plan A}), \max(\text{Plan B}), \max(\text{Plan C})\} = \max\{90, 80, 70\} = 90.$$

The Maximax choice is Plan A because it has the highest possible payoff.

$$\min(\text{Plan A}) = \min\{90, 60\} = 60,$$

$$\min(\text{Plan B}) = \min\{80, 40\} = 40,$$

$$\min(\text{Plan C}) = \min\{70, 50\} = 50.$$

$$\max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C})\} = \max\{60, 40, 50\} = 60.$$

The Maximin choice is Plan A because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Favourable market: } \max\{90, 80, 70\} = 90,$$

$$\text{Unfavourable market: } \max\{60, 40, 50\} = 60.$$

Regret table (best payoff – payoff):

| Alternative | Favourable     | Unfavourable   | Maximum regret |
|-------------|----------------|----------------|----------------|
| Plan A      | $90 - 90 = 0$  | $60 - 60 = 0$  | 0              |
| Plan B      | $90 - 80 = 10$ | $60 - 40 = 20$ | 20             |
| Plan C      | $90 - 70 = 20$ | $60 - 50 = 10$ | 20             |

The Minimax Regret choice is Plan A because it has the smallest maximum regret, 0.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Plan A         |
| Maximin        | Plan A         |
| Minimax Regret | Plan A         |



## 4 Problems $3 \times 2$

### Problems:

- 4.1. Routing System (AAB)
- 4.2. Service Model (ABA)
- 4.3. Inventory Policy (ABB)
- 4.4. Logistics Contract (AAA)

### 4.1 Routing System

A shipping company must choose between System A and System B. Three states of nature represent different operating conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a routing system.

| <i>Payoff table</i> |          |          |
|---------------------|----------|----------|
| State of nature     | System A | System B |
| State $S_1$         | 100      | 70       |
| State $S_2$         | 50       | 49       |
| State $S_3$         | 50       | 90       |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{System A}) = \max\{100, 50, 50\} = 100,$$

$$\max(\text{System B}) = \max\{70, 49, 90\} = 90.$$

$$\max\{\max(\text{System A}), \max(\text{System B})\} = \max\{100, 90\} = 100.$$

The Maximax choice is System A because it has the highest possible payoff.

$$\min(\text{System A}) = \min\{100, 50, 50\} = 50,$$

$$\min(\text{System B}) = \min\{70, 49, 90\} = 49.$$

$$\max\{\min(\text{System A}), \min(\text{System B})\} = \max\{50, 49\} = 50.$$

The Maximin choice is System A because it has the largest worst-case payoff.

## C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70\} = 100,$$

$$\text{State } S_2 : \max\{50, 49\} = 50,$$

$$\text{State } S_3 : \max\{50, 90\} = 90.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$         | $S_3$          | Maximum regret |
|-------------|-----------------|---------------|----------------|----------------|
| System A    | $100 - 100 = 0$ | $50 - 50 = 0$ | $90 - 50 = 40$ | 40             |
| System B    | $100 - 70 = 30$ | $50 - 49 = 1$ | $90 - 90 = 0$  | 30             |

The Minimax Regret choice is System B because it has the smallest maximum regret, 30.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | System A       |
| Maximin        | System A       |
| Minimax Regret | System B       |



## 4.2 Service Model

A service company must choose between Service Model A and Service Model B. Demand can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service model.

| <i>Payoff table</i> |         |         |
|---------------------|---------|---------|
| State of nature     | Model A | Model B |
| High demand         | 100     | 80      |
| Moderate demand     | 45      | 50      |
| Low demand          | 40      | 50      |

## Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Model A}) = \max\{100, 45, 40\} = 100,$$

$$\max(\text{Model B}) = \max\{80, 50, 50\} = 80.$$

$$\max\{\max(\text{Model A}), \max(\text{Model B})\} = \max\{100, 80\} = 100.$$

The Maximax choice is Model A because it has the highest possible payoff.

$$\min(\text{Model A}) = \min\{100, 45, 40\} = 40,$$

$$\min(\text{Model B}) = \min\{80, 50, 50\} = 50.$$

$$\max\{\min(\text{Model A}), \min(\text{Model B})\} = \max\{40, 50\} = 50.$$

The Maximin choice is Model B because it has the largest worst-case payoff.

### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High demand: } \max\{100, 80\} = 100,$$

$$\text{Moderate demand: } \max\{45, 50\} = 50,$$

$$\text{Low demand: } \max\{40, 50\} = 50.$$

Regret table (best payoff – payoff):

| Alternative | High            | Moderate      | Low            | Maximum regret |
|-------------|-----------------|---------------|----------------|----------------|
| Model A     | $100 - 100 = 0$ | $50 - 45 = 5$ | $50 - 40 = 10$ | 10             |
| Model B     | $100 - 80 = 20$ | $50 - 50 = 0$ | $50 - 50 = 0$  | 20             |

The Minimax Regret choice is Model A because it has the smallest maximum regret, 10.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Model A        |
| Maximin        | Model B        |
| Minimax Regret | Model A        |



### 4.3 Inventory Policy

A retailer must choose between Inventory Policy A and Inventory Policy B. Demand can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory policy.

| <i>Payoff table</i> |          |          |
|---------------------|----------|----------|
| State of nature     | Policy A | Policy B |
| High demand         | 100      | 70       |
| Moderate demand     | 20       | 60       |
| Low demand          | 0        | 50       |

#### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Policy A}) = \max\{100, 20, 0\} = 100,$$

$$\max(\text{Policy B}) = \max\{70, 60, 50\} = 70.$$

$$\max\{\max(\text{Policy A}), \max(\text{Policy B})\} = \max\{100, 70\} = 100.$$

The Maximax choice is Policy A because it has the highest possible payoff.

$$\min(\text{Policy A}) = \min\{100, 20, 0\} = 0,$$

$$\min(\text{Policy B}) = \min\{70, 60, 50\} = 50.$$

$$\max\{\min(\text{Policy A}), \min(\text{Policy B})\} = \max\{0, 50\} = 50.$$

The Maximin choice is Policy B because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High demand: } \max\{100, 70\} = 100,$$

$$\text{Moderate demand: } \max\{20, 60\} = 60,$$

$$\text{Low demand: } \max\{0, 50\} = 50.$$

Regret table (best payoff – payoff):

| Alternative | High            | Moderate       | Low           | Maximum regret |
|-------------|-----------------|----------------|---------------|----------------|
| Policy A    | $100 - 100 = 0$ | $60 - 20 = 40$ | $50 - 0 = 50$ | 50             |
| Policy B    | $100 - 70 = 30$ | $60 - 60 = 0$  | $50 - 50 = 0$ | 30             |

The Minimax Regret choice is Policy B because it has the smallest maximum regret, 30.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Policy A       |
| Maximin        | Policy B       |
| Minimax Regret | Policy B       |



### 4.4 Logistics Contract

A company must choose between Logistics Contract A and Logistics Contract B. Market conditions can be favourable, stable, or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a logistics contract.

| <i>Payoff table</i> |            |            |
|---------------------|------------|------------|
| State of nature     | Contract A | Contract B |
| Favourable market   | 90         | 80         |
| Stable market       | 70         | 50         |
| Unfavourable market | 60         | 40         |

#### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Contract A}) = \max\{90, 70, 60\} = 90,$$

$$\max(\text{Contract B}) = \max\{80, 50, 40\} = 80.$$

$$\max\{\max(\text{Contract A}), \max(\text{Contract B})\} = \max\{90, 80\} = 90.$$

The Maximax choice is Contract A because it has the highest possible payoff.

$$\min(\text{Contract A}) = \min\{90, 70, 60\} = 60,$$

$$\min(\text{Contract B}) = \min\{80, 50, 40\} = 40.$$

$$\max\{\min(\text{Contract A}), \min(\text{Contract B})\} = \max\{60, 40\} = 60.$$

The Maximin choice is Contract A because it has the largest worst-case payoff.

C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

Favourable market:  $\max\{90, 80\} = 90$ ,  
Stable market:  $\max\{70, 50\} = 70$ ,  
Unfavourable market:  $\max\{60, 40\} = 60$ .

Regret table (best payoff – payoff):

| Alternative | Favourable     | Stable         | Unfavourable   | Maximum regret |
|-------------|----------------|----------------|----------------|----------------|
| Contract A  | $90 - 90 = 0$  | $70 - 70 = 0$  | $60 - 60 = 0$  | 0              |
| Contract B  | $90 - 80 = 10$ | $70 - 50 = 20$ | $60 - 40 = 20$ | 20             |

The Minimax Regret choice is Contract A because it has the smallest maximum regret, 0.

Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Contract A     |
| Maximin        | Contract A     |
| Minimax Regret | Contract A     |



5 Problems  $3 \times 3$

| Problems:                      |
|--------------------------------|
| 5.1. Technology Platform (AAB) |
| 5.2. Service Model (ABA)       |
| 5.3. Inventory Policy (ABB)    |
| 5.4. Event Venue (ABC)         |
| 5.5. Distribution Plan (AAA)   |

5.1 Technology Platform

A company must choose Technology Platform A, Platform B, or Platform C. Operating conditions can be favourable, stable, or disrupted. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a technology platform.

*Payoff table*

| State of nature       | Platform A | Platform B | Platform C |
|-----------------------|------------|------------|------------|
| Favourable conditions | 100        | 70         | 60         |
| Stable conditions     | 50         | 49         | 40         |
| Disrupted conditions  | 50         | 90         | 45         |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Platform A}) = \max\{100, 50, 50\} = 100,$$

$$\max(\text{Platform B}) = \max\{70, 49, 90\} = 90,$$

$$\max(\text{Platform C}) = \max\{60, 40, 45\} = 60.$$

$$\max\{\max(\text{Platform A}), \max(\text{Platform B}), \max(\text{Platform C})\} = \max\{100, 90, 60\} = 100.$$

The Maximax choice is Platform A because it has the highest possible payoff.

$$\min(\text{Platform A}) = \min\{100, 50, 50\} = 50,$$

$$\min(\text{Platform B}) = \min\{70, 49, 90\} = 49,$$

$$\min(\text{Platform C}) = \min\{60, 40, 45\} = 40.$$

$$\max\{\min(\text{Platform A}), \min(\text{Platform B}), \min(\text{Platform C})\} = \max\{50, 49, 40\} = 50.$$

The Maximin choice is Platform A because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Favourable conditions: } \max\{100, 70, 60\} = 100,$$

$$\text{Stable conditions: } \max\{50, 49, 40\} = 50,$$

$$\text{Disrupted conditions: } \max\{50, 90, 45\} = 90.$$

Regret table (best payoff – payoff):

| Alternative | Favourable      | Stable         | Disrupted      | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|
| Platform A  | $100 - 100 = 0$ | $50 - 50 = 0$  | $90 - 50 = 40$ | 40             |
| Platform B  | $100 - 70 = 30$ | $50 - 49 = 1$  | $90 - 90 = 0$  | 30             |
| Platform C  | $100 - 60 = 40$ | $50 - 40 = 10$ | $90 - 45 = 45$ | 45             |

The Minimax Regret choice is Platform B because it has the smallest maximum regret, 30.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Platform A     |
| Maximin        | Platform A     |
| Minimax Regret | Platform B     |



## 5.2 Service Model

A service company must choose Service Model A, Model B, or Model C. Demand can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service model.

| <i>Payoff table</i> |         |         |         |
|---------------------|---------|---------|---------|
| State of nature     | Model A | Model B | Model C |
| High demand         | 100     | 80      | 70      |
| Moderate demand     | 45      | 50      | 35      |
| Low demand          | 40      | 50      | 45      |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Model A}) = \max\{100, 45, 40\} = 100,$$

$$\max(\text{Model B}) = \max\{80, 50, 50\} = 80,$$

$$\max(\text{Model C}) = \max\{70, 35, 45\} = 70.$$

$$\max\{\max(\text{Model A}), \max(\text{Model B}), \max(\text{Model C})\} = \max\{100, 80, 70\} = 100.$$

The Maximax choice is Model A because it has the highest possible payoff.

$$\min(\text{Model A}) = \min\{100, 45, 40\} = 40,$$

$$\min(\text{Model B}) = \min\{80, 50, 50\} = 50,$$

$$\min(\text{Model C}) = \min\{70, 35, 45\} = 35.$$

$$\max\{\min(\text{Model A}), \min(\text{Model B}), \min(\text{Model C})\} = \max\{40, 50, 35\} = 50.$$

The Maximin choice is Model B because it has the largest worst-case payoff.

## C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}\text{High demand: } \max\{100, 80, 70\} &= 100, \\ \text{Moderate demand: } \max\{45, 50, 35\} &= 50, \\ \text{Low demand: } \max\{40, 50, 45\} &= 50.\end{aligned}$$

Regret table (best payoff – payoff):

| Alternative | High            | Moderate       | Low            | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|
| Model A     | $100 - 100 = 0$ | $50 - 45 = 5$  | $50 - 40 = 10$ | 10             |
| Model B     | $100 - 80 = 20$ | $50 - 50 = 0$  | $50 - 50 = 0$  | 20             |
| Model C     | $100 - 70 = 30$ | $50 - 35 = 15$ | $50 - 45 = 5$  | 30             |

The Minimax Regret choice is Model A because it has the smallest maximum regret, 10.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Model A        |
| Maximin        | Model B        |
| Minimax Regret | Model A        |



## 5.3 Inventory Policy

A retailer must choose Inventory Policy A, Policy B, or Policy C. Demand can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory policy.

| <i>Payoff table</i> |          |          |          |
|---------------------|----------|----------|----------|
| State of nature     | Policy A | Policy B | Policy C |
| High demand         | 100      | 70       | 60       |
| Moderate demand     | 20       | 60       | 40       |
| Low demand          | 0        | 50       | 30       |

### Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Policy A}) = \max\{100, 20, 0\} = 100,$$

$$\max(\text{Policy B}) = \max\{70, 60, 50\} = 70,$$

$$\max(\text{Policy C}) = \max\{60, 40, 30\} = 60.$$

$$\max\{\max(\text{Policy A}), \max(\text{Policy B}), \max(\text{Policy C})\} = \max\{100, 70, 60\} = 100.$$

The Maximax choice is Policy A because it has the highest possible payoff.

$$\min(\text{Policy A}) = \min\{100, 20, 0\} = 0,$$

$$\min(\text{Policy B}) = \min\{70, 60, 50\} = 50,$$

$$\min(\text{Policy C}) = \min\{60, 40, 30\} = 30.$$

$$\max\{\min(\text{Policy A}), \min(\text{Policy B}), \min(\text{Policy C})\} = \max\{0, 50, 30\} = 50.$$

The Maximin choice is Policy B because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High demand: } \max\{100, 70, 60\} = 100,$$

$$\text{Moderate demand: } \max\{20, 60, 40\} = 60,$$

$$\text{Low demand: } \max\{0, 50, 30\} = 50.$$

Regret table (best payoff – payoff):

| Alternative | High            | Moderate       | Low            | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|
| Policy A    | $100 - 100 = 0$ | $60 - 20 = 40$ | $50 - 0 = 50$  | 50             |
| Policy B    | $100 - 70 = 30$ | $60 - 60 = 0$  | $50 - 50 = 0$  | 30             |
| Policy C    | $100 - 60 = 40$ | $60 - 40 = 20$ | $50 - 30 = 20$ | 40             |

The Minimax Regret choice is Policy B because it has the smallest maximum regret, 30.

#### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Policy A       |
| Maximin        | Policy B       |
| Minimax Regret | Policy B       |



## 5.4 Event Venue

An event organiser must choose Venue A, Venue B, or Venue C. Attendance can be high, moderate, or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an event venue.

| <i>Payoff table</i> |         |         |         |
|---------------------|---------|---------|---------|
| State of nature     | Venue A | Venue B | Venue C |
| High attendance     | 100     | 70      | 80      |
| Moderate attendance | 0       | 60      | 50      |
| Low attendance      | 20      | 55      | 50      |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Venue A}) = \max\{100, 0, 20\} = 100,$$

$$\max(\text{Venue B}) = \max\{70, 60, 55\} = 70,$$

$$\max(\text{Venue C}) = \max\{80, 50, 50\} = 80.$$

$$\max\{\max(\text{Venue A}), \max(\text{Venue B}), \max(\text{Venue C})\} = \max\{100, 70, 80\} = 100.$$

The Maximax choice is Venue A because it has the highest possible payoff.

$$\min(\text{Venue A}) = \min\{100, 0, 20\} = 0,$$

$$\min(\text{Venue B}) = \min\{70, 60, 55\} = 55,$$

$$\min(\text{Venue C}) = \min\{80, 50, 50\} = 50.$$

$$\max\{\min(\text{Venue A}), \min(\text{Venue B}), \min(\text{Venue C})\} = \max\{0, 55, 50\} = 55.$$

The Maximin choice is Venue B because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High attendance: } \max\{100, 70, 80\} = 100,$$

$$\text{Moderate attendance: } \max\{0, 60, 50\} = 60,$$

$$\text{Low attendance: } \max\{20, 55, 50\} = 55.$$

Regret table (best payoff – payoff):

| Alternative | High            | Moderate       | Low            | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|
| Venue A     | $100 - 100 = 0$ | $60 - 0 = 60$  | $55 - 20 = 35$ | 60             |
| Venue B     | $100 - 70 = 30$ | $60 - 60 = 0$  | $55 - 55 = 0$  | 30             |
| Venue C     | $100 - 80 = 20$ | $60 - 50 = 10$ | $55 - 50 = 5$  | 20             |

The Minimax Regret choice is Venue C because it has the smallest maximum regret, 20.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Venue A        |
| Maximin        | Venue B        |
| Minimax Regret | Venue C        |



## 5.5 Distribution Plan

A company must choose Distribution Plan A, Plan B, or Plan C. Market conditions can be favourable, stable, or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a distribution plan.

| <i>Payoff table</i> |        |        |        |
|---------------------|--------|--------|--------|
| State of nature     | Plan A | Plan B | Plan C |
| Favourable market   | 90     | 80     | 70     |
| Stable market       | 70     | 50     | 60     |
| Unfavourable market | 60     | 40     | 30     |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Plan A}) = \max\{90, 70, 60\} = 90,$$

$$\max(\text{Plan B}) = \max\{80, 50, 40\} = 80,$$

$$\max(\text{Plan C}) = \max\{70, 60, 30\} = 70.$$

$$\max\{\max(\text{Plan A}), \max(\text{Plan B}), \max(\text{Plan C})\} = \max\{90, 80, 70\} = 90.$$

The Maximax choice is Plan A because it has the highest possible payoff.

$$\min(\text{Plan A}) = \min\{90, 70, 60\} = 60,$$

$$\min(\text{Plan B}) = \min\{80, 50, 40\} = 40,$$

$$\min(\text{Plan C}) = \min\{70, 60, 30\} = 30.$$

$$\max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C})\} = \max\{60, 40, 30\} = 60.$$

The Maximin choice is Plan A because it has the largest worst-case payoff.

## C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Favourable market: } \max\{90, 80, 70\} = 90,$$

$$\text{Stable market: } \max\{70, 50, 60\} = 70,$$

$$\text{Unfavourable market: } \max\{60, 40, 30\} = 60.$$

Regret table (best payoff – payoff):

| Alternative | Favourable     | Stable         | Unfavourable   | Maximum regret |
|-------------|----------------|----------------|----------------|----------------|
| Plan A      | $90 - 90 = 0$  | $70 - 70 = 0$  | $60 - 60 = 0$  | 0              |
| Plan B      | $90 - 80 = 10$ | $70 - 50 = 20$ | $60 - 40 = 20$ | 20             |
| Plan C      | $90 - 70 = 20$ | $70 - 60 = 10$ | $60 - 30 = 30$ | 30             |

The Minimax Regret choice is Plan A because it has the smallest maximum regret, 0.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Plan A         |
| Maximin        | Plan A         |
| Minimax Regret | Plan A         |



## 6 Problems $4 \times 2$

### Problems:

6.1. Production System (AAB)

6.2. Service Network (ABA)

6.3. Inventory Strategy (ABB)

6.4. Supply Contract (AAA)

## 6.1 Production System

A manufacturing company must choose between Production System A and Production System B. Four states of nature represent different operating conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a production system.

| <i>Payoff table</i> |          |          |
|---------------------|----------|----------|
| State of nature     | System A | System B |
| State $S_1$         | 100      | 70       |
| State $S_2$         | 50       | 49       |
| State $S_3$         | 50       | 90       |
| State $S_4$         | 55       | 52       |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{System A}) = \max\{100, 50, 50, 55\} = 100,$$

$$\max(\text{System B}) = \max\{70, 49, 90, 52\} = 90.$$

$$\max\{\max(\text{System A}), \max(\text{System B})\} = \max\{100, 90\} = 100.$$

The Maximax choice is System A because it has the highest possible payoff.

$$\min(\text{System A}) = \min\{100, 50, 50, 55\} = 50,$$

$$\min(\text{System B}) = \min\{70, 49, 90, 52\} = 49.$$

$$\max\{\min(\text{System A}), \min(\text{System B})\} = \max\{50, 49\} = 50.$$

The Maximin choice is System A because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70\} = 100,$$

$$\text{State } S_2 : \max\{50, 49\} = 50,$$

$$\text{State } S_3 : \max\{50, 90\} = 90,$$

$$\text{State } S_4 : \max\{55, 52\} = 55.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$         | $S_3$          | $S_4$         | Maximum regret |
|-------------|-----------------|---------------|----------------|---------------|----------------|
| System A    | $100 - 100 = 0$ | $50 - 50 = 0$ | $90 - 50 = 40$ | $55 - 55 = 0$ | 40             |
| System B    | $100 - 70 = 30$ | $50 - 49 = 1$ | $90 - 90 = 0$  | $55 - 52 = 3$ | 30             |

The Minimax Regret choice is System B because it has the smallest maximum regret, 30.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | System A       |
| Maximin        | System A       |
| Minimax Regret | System B       |



## 6.2 Service Network

A service company must choose between Service Network A and Service Network B. Four states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service network.

| <i>Payoff table</i> |           |           |
|---------------------|-----------|-----------|
| State of nature     | Network A | Network B |
| State $S_1$         | 100       | 80        |
| State $S_2$         | 45        | 50        |
| State $S_3$         | 40        | 50        |
| State $S_4$         | 48        | 52        |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Network A}) = \max\{100, 45, 40, 48\} = 100,$$

$$\max(\text{Network B}) = \max\{80, 50, 50, 52\} = 80.$$

$$\max\{\max(\text{Network A}), \max(\text{Network B})\} = \max\{100, 80\} = 100.$$

The Maximax choice is Network A because it has the highest possible payoff.

$$\min(\text{Network A}) = \min\{100, 45, 40, 48\} = 40,$$

$$\min(\text{Network B}) = \min\{80, 50, 50, 52\} = 50.$$

$$\max\{\min(\text{Network A}), \min(\text{Network B})\} = \max\{40, 50\} = 50.$$

The Maximin choice is Network B because it has the largest worst-case payoff.

### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 80\} = 100,$$

$$\text{State } S_2 : \max\{45, 50\} = 50,$$

$$\text{State } S_3 : \max\{40, 50\} = 50,$$

$$\text{State } S_4 : \max\{48, 52\} = 52.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$         | $S_3$          | $S_4$         | Maximum regret |
|-------------|-----------------|---------------|----------------|---------------|----------------|
| Network A   | $100 - 100 = 0$ | $50 - 45 = 5$ | $50 - 40 = 10$ | $52 - 48 = 4$ | 10             |
| Network B   | $100 - 80 = 20$ | $50 - 50 = 0$ | $50 - 50 = 0$  | $52 - 52 = 0$ | 20             |

The Minimax Regret choice is Network A because it has the smallest maximum regret, 10.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Network A      |
| Maximin        | Network B      |
| Minimax Regret | Network A      |



## 6.3 Inventory Strategy

A retailer must choose between Inventory Strategy A and Inventory Strategy B. Four states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory strategy.

*Payoff table*

| State of nature | Strategy A | Strategy B |
|-----------------|------------|------------|
| State $S_1$     | 100        | 70         |
| State $S_2$     | 20         | 60         |
| State $S_3$     | 0          | 50         |
| State $S_4$     | 25         | 55         |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Strategy A}) = \max\{100, 20, 0, 25\} = 100,$$

$$\max(\text{Strategy B}) = \max\{70, 60, 50, 55\} = 70.$$

$$\max\{\max(\text{Strategy A}), \max(\text{Strategy B})\} = \max\{100, 70\} = 100.$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\min(\text{Strategy A}) = \min\{100, 20, 0, 25\} = 0,$$

$$\min(\text{Strategy B}) = \min\{70, 60, 50, 55\} = 50.$$

$$\max\{\min(\text{Strategy A}), \min(\text{Strategy B})\} = \max\{0, 50\} = 50.$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70\} = 100,$$

$$\text{State } S_2 : \max\{20, 60\} = 60,$$

$$\text{State } S_3 : \max\{0, 50\} = 50,$$

$$\text{State } S_4 : \max\{25, 55\} = 55.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$         | $S_4$          | Maximum regret |
|-------------|-----------------|----------------|---------------|----------------|----------------|
| Strategy A  | $100 - 100 = 0$ | $60 - 20 = 40$ | $50 - 0 = 50$ | $55 - 25 = 30$ | 50             |
| Strategy B  | $100 - 70 = 30$ | $60 - 60 = 0$  | $50 - 50 = 0$ | $55 - 55 = 0$  | 30             |

The Minimax Regret choice is Strategy B because it has the smallest maximum regret, 30.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Strategy A     |
| Maximin        | Strategy B     |
| Minimax Regret | Strategy B     |



## 6.4 Supply Contract

A company must choose between Supply Contract A and Supply Contract B. Four states of nature represent different market and supply conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a supply contract.

| <i>Payoff table</i> |            |            |
|---------------------|------------|------------|
| State of nature     | Contract A | Contract B |
| State $S_1$         | 90         | 80         |
| State $S_2$         | 70         | 50         |
| State $S_3$         | 60         | 40         |
| State $S_4$         | 65         | 55         |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Contract A}) = \max\{90, 70, 60, 65\} = 90,$$

$$\max(\text{Contract B}) = \max\{80, 50, 40, 55\} = 80.$$

$$\max\{\max(\text{Contract A}), \max(\text{Contract B})\} = \max\{90, 80\} = 90.$$

The Maximax choice is Contract A because it has the highest possible payoff.

$$\min(\text{Contract A}) = \min\{90, 70, 60, 65\} = 60,$$

$$\min(\text{Contract B}) = \min\{80, 50, 40, 55\} = 40.$$

$$\max\{\min(\text{Contract A}), \min(\text{Contract B})\} = \max\{60, 40\} = 60.$$

The Maximin choice is Contract A because it has the largest worst-case payoff.

## C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{90, 80\} = 90,$$

$$\text{State } S_2 : \max\{70, 50\} = 70,$$

$$\text{State } S_3 : \max\{60, 40\} = 60,$$

$$\text{State } S_4 : \max\{65, 55\} = 65.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$          | $S_2$          | $S_3$          | $S_4$          | Maximum regret |
|-------------|----------------|----------------|----------------|----------------|----------------|
| Contract A  | $90 - 90 = 0$  | $70 - 70 = 0$  | $60 - 60 = 0$  | $65 - 65 = 0$  | 0              |
| Contract B  | $90 - 80 = 10$ | $70 - 50 = 20$ | $60 - 40 = 20$ | $65 - 55 = 10$ | 20             |

The Minimax Regret choice is Contract A because it has the smallest maximum regret, 0.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Contract A     |
| Maximin        | Contract A     |
| Minimax Regret | Contract A     |



## 7 Problems $2 \times 4$

### Problems:

7.1. Market Entry Strategy (ABA)

7.2. Production Plan (ABB)

7.3. Facility Location (ABC)

7.4. Supply Agreement (AAA)

### 7.1 Market Entry Strategy

A company must choose among Market Entry Strategy A, Strategy B, Strategy C, and Strategy D. Market conditions can be favourable or unfavourable. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a market entry strategy.

| <i>Payoff table</i> |            |            |            |            |
|---------------------|------------|------------|------------|------------|
| State of nature     | Strategy A | Strategy B | Strategy C | Strategy D |
| Favourable market   | 30         | 60         | 50         | 0          |
| Unfavourable market | 100        | 40         | 10         | 30         |

## Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Strategy A}) = \max\{30, 100\} = 100,$$

$$\max(\text{Strategy B}) = \max\{60, 40\} = 60,$$

$$\max(\text{Strategy C}) = \max\{50, 10\} = 50,$$

$$\max(\text{Strategy D}) = \max\{0, 30\} = 30.$$

$$\begin{aligned} &\max\{\max(\text{Strategy A}), \max(\text{Strategy B}), \max(\text{Strategy C}), \max(\text{Strategy D})\} \\ &= \max\{100, 60, 50, 30\} = 100. \end{aligned}$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\min(\text{Strategy A}) = \min\{30, 100\} = 30,$$

$$\min(\text{Strategy B}) = \min\{60, 40\} = 40,$$

$$\min(\text{Strategy C}) = \min\{50, 10\} = 10,$$

$$\min(\text{Strategy D}) = \min\{0, 30\} = 0.$$

$$\begin{aligned} &\max\{\min(\text{Strategy A}), \min(\text{Strategy B}), \min(\text{Strategy C}), \min(\text{Strategy D})\} \\ &= \max\{30, 40, 10, 0\} = 40. \end{aligned}$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Favourable market: } \max\{30, 60, 50, 0\} = 60,$$

$$\text{Unfavourable market: } \max\{100, 40, 10, 30\} = 100.$$

Regret table (best payoff – payoff):

| Alternative | Favourable     | Unfavourable    | Maximum regret |
|-------------|----------------|-----------------|----------------|
| Strategy A  | $60 - 30 = 30$ | $100 - 100 = 0$ | 30             |
| Strategy B  | $60 - 60 = 0$  | $100 - 40 = 60$ | 60             |
| Strategy C  | $60 - 50 = 10$ | $100 - 10 = 90$ | 90             |
| Strategy D  | $60 - 0 = 60$  | $100 - 30 = 70$ | 70             |

The Minimax Regret choice is Strategy A because it has the smallest maximum regret, 30.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Strategy A     |
| Maximin        | Strategy B     |
| Minimax Regret | Strategy A     |



## 7.2 Production Plan

A manufacturing company must choose among Production Plan A, Plan B, Plan C, and Plan D. Demand can be high or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a production plan.

| <i>Payoff table</i> |        |        |        |        |
|---------------------|--------|--------|--------|--------|
| State of nature     | Plan A | Plan B | Plan C | Plan D |
| High demand         | 100    | 90     | 40     | 30     |
| Low demand          | 70     | 90     | 80     | 40     |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Plan A}) = \max\{100, 70\} = 100,$$

$$\max(\text{Plan B}) = \max\{90, 90\} = 90,$$

$$\max(\text{Plan C}) = \max\{40, 80\} = 80,$$

$$\max(\text{Plan D}) = \max\{30, 40\} = 40.$$

$$\max\{\max(\text{Plan A}), \max(\text{Plan B}), \max(\text{Plan C}), \max(\text{Plan D})\}$$

$$= \max\{100, 90, 80, 40\} = 100.$$

The Maximax choice is Plan A because it has the highest possible payoff.

$$\min(\text{Plan A}) = \min\{100, 70\} = 70,$$

$$\min(\text{Plan B}) = \min\{90, 90\} = 90,$$

$$\min(\text{Plan C}) = \min\{40, 80\} = 40,$$

$$\min(\text{Plan D}) = \min\{30, 40\} = 30.$$

$$\begin{aligned} & \max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C}), \min(\text{Plan D})\} \\ & = \max\{70, 90, 40, 30\} = 90. \end{aligned}$$

The Maximin choice is Plan B because it has the largest worst-case payoff.

### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{High demand: } \max\{100, 90, 40, 30\} = 100,$$

$$\text{Low demand: } \max\{70, 90, 80, 40\} = 90.$$

Regret table (best payoff – payoff):

| Alternative | High demand     | Low demand     | Maximum regret |
|-------------|-----------------|----------------|----------------|
| Plan A      | $100 - 100 = 0$ | $90 - 70 = 20$ | 20             |
| Plan B      | $100 - 90 = 10$ | $90 - 90 = 0$  | 10             |
| Plan C      | $100 - 40 = 60$ | $90 - 80 = 10$ | 60             |
| Plan D      | $100 - 30 = 70$ | $90 - 40 = 50$ | 70             |

The Minimax Regret choice is Plan B because it has the smallest maximum regret, 10.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Plan A         |
| Maximin        | Plan B         |
| Minimax Regret | Plan B         |



## 7.3 Facility Location

A company must choose among Facility Location A, Location B, Location C, and Location D. Regional demand can be high or low. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a facility location.

| <i>Payoff table</i>  |            |            |            |            |
|----------------------|------------|------------|------------|------------|
| State of nature      | Location A | Location B | Location C | Location D |
| High regional demand | 10         | 60         | 50         | 20         |
| Low regional demand  | 100        | 70         | 80         | 40         |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

**C4 - Applies the Maximin and Maximax criteria**

$$\max(\text{Location A}) = \max\{10, 100\} = 100,$$

$$\max(\text{Location B}) = \max\{60, 70\} = 70,$$

$$\max(\text{Location C}) = \max\{50, 80\} = 80,$$

$$\max(\text{Location D}) = \max\{20, 40\} = 40.$$

$$\begin{aligned} &\max\{\max(\text{Location A}), \max(\text{Location B}), \max(\text{Location C}), \max(\text{Location D})\} \\ &= \max\{100, 70, 80, 40\} = 100. \end{aligned}$$

The Maximax choice is Location A because it has the highest possible payoff.

$$\min(\text{Location A}) = \min\{10, 100\} = 10,$$

$$\min(\text{Location B}) = \min\{60, 70\} = 60,$$

$$\min(\text{Location C}) = \min\{50, 80\} = 50,$$

$$\min(\text{Location D}) = \min\{20, 40\} = 20.$$

$$\begin{aligned} &\max\{\min(\text{Location A}), \min(\text{Location B}), \min(\text{Location C}), \min(\text{Location D})\} \\ &= \max\{10, 60, 50, 20\} = 60. \end{aligned}$$

The Maximin choice is Location B because it has the largest worst-case payoff.

**C5 - Constructs opportunity loss tables and applies the Minimax regret criterion**

Best payoff in each state:

$$\text{High regional demand: } \max\{10, 60, 50, 20\} = 60,$$

$$\text{Low regional demand: } \max\{100, 70, 80, 40\} = 100.$$

Regret table (best payoff – payoff):

| Alternative | High demand    | Low demand      | Maximum regret |
|-------------|----------------|-----------------|----------------|
| Location A  | $60 - 10 = 50$ | $100 - 100 = 0$ | 50             |
| Location B  | $60 - 60 = 0$  | $100 - 70 = 30$ | 30             |
| Location C  | $60 - 50 = 10$ | $100 - 80 = 20$ | 20             |
| Location D  | $60 - 20 = 40$ | $100 - 40 = 60$ | 60             |

The Minimax Regret choice is Location C because it has the smallest maximum regret, 20.

## Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Location A     |
| Maximin        | Location B     |
| Minimax Regret | Location C     |



## 7.4 Supply Agreement

A company must choose among Supply Agreement A, Agreement B, Agreement C, and Agreement D. Supply conditions can be stable or disrupted. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a supply agreement.

| <i>Payoff table</i> |             |             |             |             |
|---------------------|-------------|-------------|-------------|-------------|
| State of nature     | Agreement A | Agreement B | Agreement C | Agreement D |
| Stable supply       | 90          | 70          | 60          | 30          |
| Disrupted supply    | 100         | 80          | 50          | 70          |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Agreement A}) = \max\{90, 100\} = 100,$$

$$\max(\text{Agreement B}) = \max\{70, 80\} = 80,$$

$$\max(\text{Agreement C}) = \max\{60, 50\} = 60,$$

$$\max(\text{Agreement D}) = \max\{30, 70\} = 70.$$

$$\begin{aligned} & \max\{\max(\text{Agreement A}), \max(\text{Agreement B}), \max(\text{Agreement C}), \max(\text{Agreement D})\} \\ &= \max\{100, 80, 60, 70\} = 100. \end{aligned}$$

The Maximax choice is Agreement A because it has the highest possible payoff.

$$\min(\text{Agreement A}) = \min\{90, 100\} = 90,$$

$$\min(\text{Agreement B}) = \min\{70, 80\} = 70,$$

$$\min(\text{Agreement C}) = \min\{60, 50\} = 50,$$

$$\min(\text{Agreement D}) = \min\{30, 70\} = 30.$$

$$\begin{aligned} & \max\{\min(\text{Agreement A}), \min(\text{Agreement B}), \min(\text{Agreement C}), \min(\text{Agreement D})\} \\ & = \max\{90, 70, 50, 30\} = 90. \end{aligned}$$

The Maximin choice is Agreement A because it has the largest worst-case payoff.

### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{Stable supply: } \max\{90, 70, 60, 30\} = 90,$$

$$\text{Disrupted supply: } \max\{100, 80, 50, 70\} = 100.$$

Regret table (best payoff – payoff):

| Alternative | Stable supply  | Disrupted supply | Maximum regret |
|-------------|----------------|------------------|----------------|
| Agreement A | $90 - 90 = 0$  | $100 - 100 = 0$  | 0              |
| Agreement B | $90 - 70 = 20$ | $100 - 80 = 20$  | 20             |
| Agreement C | $90 - 60 = 30$ | $100 - 50 = 50$  | 50             |
| Agreement D | $90 - 30 = 60$ | $100 - 70 = 30$  | 60             |

The Minimax Regret choice is Agreement A because it has the smallest maximum regret, 0.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Agreement A    |
| Maximin        | Agreement A    |
| Minimax Regret | Agreement A    |



## 8 Problems $4 \times 3$

### Problems:

8.1. Production System (AAB)

8.2. Service Network (ABA)

8.3. Inventory Strategy (ABB)

8.4. Event Venue (ABC)

8.5. Supply Contract (AAA)

### 8.1 Production System

A manufacturing company must choose among Production System A, System B, and System C. Four states of nature represent different operating conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a production system.

| <i>Payoff table</i> |          |          |          |
|---------------------|----------|----------|----------|
| State of nature     | System A | System B | System C |
| State $S_1$         | 100      | 70       | 60       |
| State $S_2$         | 50       | 49       | 30       |
| State $S_3$         | 50       | 90       | 40       |
| State $S_4$         | 55       | 52       | 35       |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{System A}) = \max\{100, 50, 50, 55\} = 100,$$

$$\max(\text{System B}) = \max\{70, 49, 90, 52\} = 90,$$

$$\max(\text{System C}) = \max\{60, 30, 40, 35\} = 60.$$

$$\begin{aligned} &\max\{\max(\text{System A}), \max(\text{System B}), \max(\text{System C})\} \\ &= \max\{100, 90, 60\} = 100. \end{aligned}$$

The Maximax choice is System A because it has the highest possible payoff.

$$\min(\text{System A}) = \min\{100, 50, 50, 55\} = 50,$$

$$\min(\text{System B}) = \min\{70, 49, 90, 52\} = 49,$$

$$\min(\text{System C}) = \min\{60, 30, 40, 35\} = 30.$$

$$\begin{aligned} &\max\{\min(\text{System A}), \min(\text{System B}), \min(\text{System C})\} \\ &= \max\{50, 49, 30\} = 50. \end{aligned}$$

The Maximin choice is System A because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70, 60\} = 100,$$

$$\text{State } S_2 : \max\{50, 49, 30\} = 50,$$

$$\text{State } S_3 : \max\{50, 90, 40\} = 90,$$

$$\text{State } S_4 : \max\{55, 52, 35\} = 55.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$          | $S_4$          | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|----------------|
| System A    | $100 - 100 = 0$ | $50 - 50 = 0$  | $90 - 50 = 40$ | $55 - 55 = 0$  | 40             |
| System B    | $100 - 70 = 30$ | $50 - 49 = 1$  | $90 - 90 = 0$  | $55 - 52 = 3$  | 30             |
| System C    | $100 - 60 = 40$ | $50 - 30 = 20$ | $90 - 40 = 50$ | $55 - 35 = 20$ | 50             |

The Minimax Regret choice is System B because it has the smallest maximum regret, 30.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | System A       |
| Maximin        | System A       |
| Minimax Regret | System B       |



## 8.2 Service Network

A service company must choose among Service Network A, Network B, and Network C. Four states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service network.

| <i>Payoff table</i> |           |           |           |
|---------------------|-----------|-----------|-----------|
| State of nature     | Network A | Network B | Network C |
| State $S_1$         | 100       | 80        | 60        |
| State $S_2$         | 45        | 50        | 30        |
| State $S_3$         | 40        | 50        | 35        |
| State $S_4$         | 48        | 52        | 40        |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Network A}) = \max\{100, 45, 40, 48\} = 100,$$

$$\max(\text{Network B}) = \max\{80, 50, 50, 52\} = 80,$$

$$\max(\text{Network C}) = \max\{60, 30, 35, 40\} = 60.$$

$$\begin{aligned} & \max\{\max(\text{Network A}), \max(\text{Network B}), \max(\text{Network C})\} \\ & = \max\{100, 80, 60\} = 100. \end{aligned}$$

The Maximax choice is Network A because it has the highest possible payoff.

$$\begin{aligned} \min(\text{Network A}) &= \min\{100, 45, 40, 48\} = 40, \\ \min(\text{Network B}) &= \min\{80, 50, 50, 52\} = 50, \\ \min(\text{Network C}) &= \min\{60, 30, 35, 40\} = 30. \end{aligned}$$

$$\begin{aligned} & \max\{\min(\text{Network A}), \min(\text{Network B}), \min(\text{Network C})\} \\ & = \max\{40, 50, 30\} = 50. \end{aligned}$$

The Maximin choice is Network B because it has the largest worst-case payoff.

### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned} \text{State } S_1 &: \max\{100, 80, 60\} = 100, \\ \text{State } S_2 &: \max\{45, 50, 30\} = 50, \\ \text{State } S_3 &: \max\{40, 50, 35\} = 50, \\ \text{State } S_4 &: \max\{48, 52, 40\} = 52. \end{aligned}$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$          | $S_4$          | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|----------------|
| Network A   | $100 - 100 = 0$ | $50 - 45 = 5$  | $50 - 40 = 10$ | $52 - 48 = 4$  | 10             |
| Network B   | $100 - 80 = 20$ | $50 - 50 = 0$  | $50 - 50 = 0$  | $52 - 52 = 0$  | 20             |
| Network C   | $100 - 60 = 40$ | $50 - 30 = 20$ | $50 - 35 = 15$ | $52 - 40 = 12$ | 40             |

The Minimax Regret choice is Network A because it has the smallest maximum regret, 10.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Network A      |
| Maximin        | Network B      |
| Minimax Regret | Network A      |



## 8.3 Inventory Strategy

A retailer must choose among Inventory Strategy A, Strategy B, and Strategy C. Four states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory strategy.

| <i>Payoff table</i> |            |            |            |
|---------------------|------------|------------|------------|
| State of nature     | Strategy A | Strategy B | Strategy C |
| State $S_1$         | 100        | 70         | 50         |
| State $S_2$         | 20         | 60         | 40         |
| State $S_3$         | 0          | 50         | 30         |
| State $S_4$         | 25         | 55         | 45         |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Strategy A}) = \max\{100, 20, 0, 25\} = 100,$$

$$\max(\text{Strategy B}) = \max\{70, 60, 50, 55\} = 70,$$

$$\max(\text{Strategy C}) = \max\{50, 40, 30, 45\} = 50.$$

$$\begin{aligned} \max\{\max(\text{Strategy A}), \max(\text{Strategy B}), \max(\text{Strategy C})\} \\ = \max\{100, 70, 50\} = 100. \end{aligned}$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\min(\text{Strategy A}) = \min\{100, 20, 0, 25\} = 0,$$

$$\min(\text{Strategy B}) = \min\{70, 60, 50, 55\} = 50,$$

$$\min(\text{Strategy C}) = \min\{50, 40, 30, 45\} = 30.$$

$$\begin{aligned} \max\{\min(\text{Strategy A}), \min(\text{Strategy B}), \min(\text{Strategy C})\} \\ = \max\{0, 50, 30\} = 50. \end{aligned}$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70, 50\} = 100,$$

$$\text{State } S_2 : \max\{20, 60, 40\} = 60,$$

$$\text{State } S_3 : \max\{0, 50, 30\} = 50,$$

$$\text{State } S_4 : \max\{25, 55, 45\} = 55.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$          | $S_4$          | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|----------------|
| Strategy A  | $100 - 100 = 0$ | $60 - 20 = 40$ | $50 - 0 = 50$  | $55 - 25 = 30$ | 50             |
| Strategy B  | $100 - 70 = 30$ | $60 - 60 = 0$  | $50 - 50 = 0$  | $55 - 55 = 0$  | 30             |
| Strategy C  | $100 - 50 = 50$ | $60 - 40 = 20$ | $50 - 30 = 20$ | $55 - 45 = 10$ | 50             |

The Minimax Regret choice is Strategy B because it has the smallest maximum regret, 30.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Strategy A     |
| Maximin        | Strategy B     |
| Minimax Regret | Strategy B     |



## 8.4 Event Venue

An event company must choose among Venue A, Venue B, and Venue C. Four states of nature represent different attendance levels. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an event venue.

| <i>Payoff table</i> |         |         |         |
|---------------------|---------|---------|---------|
| State of nature     | Venue A | Venue B | Venue C |
| State $S_1$         | 100     | 70      | 80      |
| State $S_2$         | 0       | 60      | 50      |
| State $S_3$         | 20      | 55      | 50      |
| State $S_4$         | 40      | 58      | 52      |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Venue A}) = \max\{100, 0, 20, 40\} = 100,$$

$$\max(\text{Venue B}) = \max\{70, 60, 55, 58\} = 70,$$

$$\max(\text{Venue C}) = \max\{80, 50, 50, 52\} = 80.$$

$$\begin{aligned} & \max\{\max(\text{Venue A}), \max(\text{Venue B}), \max(\text{Venue C})\} \\ & = \max\{100, 70, 80\} = 100. \end{aligned}$$

The Maximax choice is Venue A because it has the highest possible payoff.

$$\begin{aligned} \min(\text{Venue A}) &= \min\{100, 0, 20, 40\} = 0, \\ \min(\text{Venue B}) &= \min\{70, 60, 55, 58\} = 55, \\ \min(\text{Venue C}) &= \min\{80, 50, 50, 52\} = 50. \end{aligned}$$

$$\begin{aligned} & \max\{\min(\text{Venue A}), \min(\text{Venue B}), \min(\text{Venue C})\} \\ & = \max\{0, 55, 50\} = 55. \end{aligned}$$

The Maximin choice is Venue B because it has the largest worst-case payoff.

### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned} \text{State } S_1 &: \max\{100, 70, 80\} = 100, \\ \text{State } S_2 &: \max\{0, 60, 50\} = 60, \\ \text{State } S_3 &: \max\{20, 55, 50\} = 55, \\ \text{State } S_4 &: \max\{40, 58, 52\} = 58. \end{aligned}$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$          | $S_4$          | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|----------------|
| Venue A     | $100 - 100 = 0$ | $60 - 0 = 60$  | $55 - 20 = 35$ | $58 - 40 = 18$ | 60             |
| Venue B     | $100 - 70 = 30$ | $60 - 60 = 0$  | $55 - 55 = 0$  | $58 - 58 = 0$  | 30             |
| Venue C     | $100 - 80 = 20$ | $60 - 50 = 10$ | $55 - 50 = 5$  | $58 - 52 = 6$  | 20             |

The Minimax Regret choice is Venue C because it has the smallest maximum regret, 20.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Venue A        |
| Maximin        | Venue B        |
| Minimax Regret | Venue C        |



## 8.5 Supply Contract

A company must choose among Supply Contract A, Contract B, and Contract C. Four states of nature represent different market and supply conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a supply contract.

| <i>Payoff table</i> |            |            |            |
|---------------------|------------|------------|------------|
| State of nature     | Contract A | Contract B | Contract C |
| State $S_1$         | 90         | 80         | 70         |
| State $S_2$         | 70         | 50         | 45         |
| State $S_3$         | 60         | 40         | 35         |
| State $S_4$         | 65         | 55         | 50         |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Contract A}) = \max\{90, 70, 60, 65\} = 90,$$

$$\max(\text{Contract B}) = \max\{80, 50, 40, 55\} = 80,$$

$$\max(\text{Contract C}) = \max\{70, 45, 35, 50\} = 70.$$

$$\begin{aligned} &\max\{\max(\text{Contract A}), \max(\text{Contract B}), \max(\text{Contract C})\} \\ &= \max\{90, 80, 70\} = 90. \end{aligned}$$

The Maximax choice is Contract A because it has the highest possible payoff.

$$\min(\text{Contract A}) = \min\{90, 70, 60, 65\} = 60,$$

$$\min(\text{Contract B}) = \min\{80, 50, 40, 55\} = 40,$$

$$\min(\text{Contract C}) = \min\{70, 45, 35, 50\} = 35.$$

$$\begin{aligned} &\max\{\min(\text{Contract A}), \min(\text{Contract B}), \min(\text{Contract C})\} \\ &= \max\{60, 40, 35\} = 60. \end{aligned}$$

The Maximin choice is Contract A because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{90, 80, 70\} = 90,$$

$$\text{State } S_2 : \max\{70, 50, 45\} = 70,$$

$$\text{State } S_3 : \max\{60, 40, 35\} = 60,$$

$$\text{State } S_4 : \max\{65, 55, 50\} = 65.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$          | $S_2$          | $S_3$          | $S_4$          | Maximum regret |
|-------------|----------------|----------------|----------------|----------------|----------------|
| Contract A  | $90 - 90 = 0$  | $70 - 70 = 0$  | $60 - 60 = 0$  | $65 - 65 = 0$  | 0              |
| Contract B  | $90 - 80 = 10$ | $70 - 50 = 20$ | $60 - 40 = 20$ | $65 - 55 = 10$ | 20             |
| Contract C  | $90 - 70 = 20$ | $70 - 45 = 25$ | $60 - 35 = 25$ | $65 - 50 = 15$ | 25             |

The Minimax Regret choice is Contract A because it has the smallest maximum regret, 0.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Contract A     |
| Maximin        | Contract A     |
| Minimax Regret | Contract A     |



## 9 Problems $3 \times 4$

### Problems:

- 9.1. Production System (AAB)
- 9.2. Service Network (ABA)
- 9.3. Inventory Strategy (ABB)
- 9.4. Event Venue (ABC)
- 9.5. Supply Contract (AAA)

### 9.1 Production System

A manufacturing company must choose among Production System A, System B, System C, and System D. Three states of nature represent different operating conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a production system.

| <i>Payoff table</i> |          |          |          |          |
|---------------------|----------|----------|----------|----------|
| State of nature     | System A | System B | System C | System D |
| State $S_1$         | 100      | 70       | 60       | 55       |
| State $S_2$         | 50       | 49       | 30       | 35       |
| State $S_3$         | 50       | 90       | 40       | 45       |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{System A}) = \max\{100, 50, 50\} = 100,$$

$$\max(\text{System B}) = \max\{70, 49, 90\} = 90,$$

$$\max(\text{System C}) = \max\{60, 30, 40\} = 60,$$

$$\max(\text{System D}) = \max\{55, 35, 45\} = 55.$$

$$\begin{aligned} &\max\{\max(\text{System A}), \max(\text{System B}), \max(\text{System C}), \max(\text{System D})\} \\ &= \max\{100, 90, 60, 55\} = 100. \end{aligned}$$

The Maximax choice is System A because it has the highest possible payoff.

$$\min(\text{System A}) = \min\{100, 50, 50\} = 50,$$

$$\min(\text{System B}) = \min\{70, 49, 90\} = 49,$$

$$\min(\text{System C}) = \min\{60, 30, 40\} = 30,$$

$$\min(\text{System D}) = \min\{55, 35, 45\} = 35.$$

$$\begin{aligned} &\max\{\min(\text{System A}), \min(\text{System B}), \min(\text{System C}), \min(\text{System D})\} \\ &= \max\{50, 49, 30, 35\} = 50. \end{aligned}$$

The Maximin choice is System A because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70, 60, 55\} = 100,$$

$$\text{State } S_2 : \max\{50, 49, 30, 35\} = 50,$$

$$\text{State } S_3 : \max\{50, 90, 40, 45\} = 90.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$          | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|
| System A    | $100 - 100 = 0$ | $50 - 50 = 0$  | $90 - 50 = 40$ | 40             |
| System B    | $100 - 70 = 30$ | $50 - 49 = 1$  | $90 - 90 = 0$  | 30             |
| System C    | $100 - 60 = 40$ | $50 - 30 = 20$ | $90 - 40 = 50$ | 50             |
| System D    | $100 - 55 = 45$ | $50 - 35 = 15$ | $90 - 45 = 45$ | 45             |

The Minimax Regret choice is System B because it has the smallest maximum regret, 30.

#### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | System A       |
| Maximin        | System A       |
| Minimax Regret | System B       |



## 9.2 Service Network

A service company must choose among Service Network A, Network B, Network C, and Network D. Three states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a service network.

| <i>Payoff table</i> |           |           |           |           |
|---------------------|-----------|-----------|-----------|-----------|
| State of nature     | Network A | Network B | Network C | Network D |
| State $S_1$         | 100       | 80        | 60        | 70        |
| State $S_2$         | 45        | 50        | 30        | 40        |
| State $S_3$         | 40        | 50        | 35        | 45        |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Network A}) = \max\{100, 45, 40\} = 100,$$

$$\max(\text{Network B}) = \max\{80, 50, 50\} = 80,$$

$$\max(\text{Network C}) = \max\{60, 30, 35\} = 60,$$

$$\max(\text{Network D}) = \max\{70, 40, 45\} = 70.$$

$$\begin{aligned} &\max\{\max(\text{Network A}), \max(\text{Network B}), \max(\text{Network C}), \max(\text{Network D})\} \\ &= \max\{100, 80, 60, 70\} = 100. \end{aligned}$$

The Maximax choice is Network A because it has the highest possible payoff.

$$\min(\text{Network A}) = \min\{100, 45, 40\} = 40,$$

$$\min(\text{Network B}) = \min\{80, 50, 50\} = 50,$$

$$\min(\text{Network C}) = \min\{60, 30, 35\} = 30,$$

$$\min(\text{Network D}) = \min\{70, 40, 45\} = 40.$$

$$\begin{aligned} &\max\{\min(\text{Network A}), \min(\text{Network B}), \min(\text{Network C}), \min(\text{Network D})\} \\ &= \max\{40, 50, 30, 40\} = 50. \end{aligned}$$

The Maximin choice is Network B because it has the largest worst-case payoff.

## C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 80, 60, 70\} = 100,$$

$$\text{State } S_2 : \max\{45, 50, 30, 40\} = 50,$$

$$\text{State } S_3 : \max\{40, 50, 35, 45\} = 50.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$          | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|
| Network A   | $100 - 100 = 0$ | $50 - 45 = 5$  | $50 - 40 = 10$ | 10             |
| Network B   | $100 - 80 = 20$ | $50 - 50 = 0$  | $50 - 50 = 0$  | 20             |
| Network C   | $100 - 60 = 40$ | $50 - 30 = 20$ | $50 - 35 = 15$ | 40             |
| Network D   | $100 - 70 = 30$ | $50 - 40 = 10$ | $50 - 45 = 5$  | 30             |

The Minimax Regret choice is Network A because it has the smallest maximum regret, 10.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Network A      |
| Maximin        | Network B      |
| Minimax Regret | Network A      |



## 9.3 Inventory Strategy

A retailer must choose among Inventory Strategy A, Strategy B, Strategy C, and Strategy D. Three states of nature represent different demand conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an inventory strategy.

| <i>Payoff table</i> |            |            |            |            |
|---------------------|------------|------------|------------|------------|
| State of nature     | Strategy A | Strategy B | Strategy C | Strategy D |
| State $S_1$         | 100        | 70         | 50         | 60         |
| State $S_2$         | 20         | 60         | 40         | 45         |
| State $S_3$         | 0          | 50         | 30         | 35         |

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Strategy A}) = \max\{100, 20, 0\} = 100,$$

$$\max(\text{Strategy B}) = \max\{70, 60, 50\} = 70,$$

$$\max(\text{Strategy C}) = \max\{50, 40, 30\} = 50,$$

$$\max(\text{Strategy D}) = \max\{60, 45, 35\} = 60.$$

$$\begin{aligned} &\max\{\max(\text{Strategy A}), \max(\text{Strategy B}), \max(\text{Strategy C}), \max(\text{Strategy D})\} \\ &= \max\{100, 70, 50, 60\} = 100. \end{aligned}$$

The Maximax choice is Strategy A because it has the highest possible payoff.

$$\min(\text{Strategy A}) = \min\{100, 20, 0\} = 0,$$

$$\min(\text{Strategy B}) = \min\{70, 60, 50\} = 50,$$

$$\min(\text{Strategy C}) = \min\{50, 40, 30\} = 30,$$

$$\min(\text{Strategy D}) = \min\{60, 45, 35\} = 35.$$

$$\begin{aligned} &\max\{\min(\text{Strategy A}), \min(\text{Strategy B}), \min(\text{Strategy C}), \min(\text{Strategy D})\} \\ &= \max\{0, 50, 30, 35\} = 50. \end{aligned}$$

The Maximin choice is Strategy B because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70, 50, 60\} = 100,$$

$$\text{State } S_2 : \max\{20, 60, 40, 45\} = 60,$$

$$\text{State } S_3 : \max\{0, 50, 30, 35\} = 50.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$          | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|
| Strategy A  | $100 - 100 = 0$ | $60 - 20 = 40$ | $50 - 0 = 50$  | 50             |
| Strategy B  | $100 - 70 = 30$ | $60 - 60 = 0$  | $50 - 50 = 0$  | 30             |
| Strategy C  | $100 - 50 = 50$ | $60 - 40 = 20$ | $50 - 30 = 20$ | 50             |
| Strategy D  | $100 - 60 = 40$ | $60 - 45 = 15$ | $50 - 35 = 15$ | 40             |

The Minimax Regret choice is Strategy B because it has the smallest maximum regret, 30.

#### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Strategy A     |
| Maximin        | Strategy B     |
| Minimax Regret | Strategy B     |



## 9.4 Event Venue

An event company must choose among Venue A, Venue B, Venue C, and Venue D. Three states of nature represent different attendance levels. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select an event venue.

| <i>Payoff table</i> |         |         |         |         |
|---------------------|---------|---------|---------|---------|
| State of nature     | Venue A | Venue B | Venue C | Venue D |
| State $S_1$         | 100     | 70      | 80      | 60      |
| State $S_2$         | 0       | 60      | 50      | 45      |
| State $S_3$         | 20      | 55      | 50      | 40      |

### Questions/Tasks.

**C4** Applies the Maximin and Maximax criteria

**C5** Constructs opportunity loss tables and applies the Minimax regret criterion

### C4 - Applies the Maximin and Maximax criteria

$$\max(\text{Venue A}) = \max\{100, 0, 20\} = 100,$$

$$\max(\text{Venue B}) = \max\{70, 60, 55\} = 70,$$

$$\max(\text{Venue C}) = \max\{80, 50, 50\} = 80,$$

$$\max(\text{Venue D}) = \max\{60, 45, 40\} = 60.$$

$$\begin{aligned} &\max\{\max(\text{Venue A}), \max(\text{Venue B}), \max(\text{Venue C}), \max(\text{Venue D})\} \\ &= \max\{100, 70, 80, 60\} = 100. \end{aligned}$$

The Maximax choice is Venue A because it has the highest possible payoff.

$$\min(\text{Venue A}) = \min\{100, 0, 20\} = 0,$$

$$\min(\text{Venue B}) = \min\{70, 60, 55\} = 55,$$

$$\min(\text{Venue C}) = \min\{80, 50, 50\} = 50,$$

$$\min(\text{Venue D}) = \min\{60, 45, 40\} = 40.$$

$$\begin{aligned} &\max\{\min(\text{Venue A}), \min(\text{Venue B}), \min(\text{Venue C}), \min(\text{Venue D})\} \\ &= \max\{0, 55, 50, 40\} = 55. \end{aligned}$$

The Maximin choice is Venue B because it has the largest worst-case payoff.

## C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\text{State } S_1 : \max\{100, 70, 80, 60\} = 100,$$

$$\text{State } S_2 : \max\{0, 60, 50, 45\} = 60,$$

$$\text{State } S_3 : \max\{20, 55, 50, 40\} = 55.$$

Regret table (best payoff – payoff):

| Alternative | $S_1$           | $S_2$          | $S_3$          | Maximum regret |
|-------------|-----------------|----------------|----------------|----------------|
| Venue A     | $100 - 100 = 0$ | $60 - 0 = 60$  | $55 - 20 = 35$ | 60             |
| Venue B     | $100 - 70 = 30$ | $60 - 60 = 0$  | $55 - 55 = 0$  | 30             |
| Venue C     | $100 - 80 = 20$ | $60 - 50 = 10$ | $55 - 50 = 5$  | 20             |
| Venue D     | $100 - 60 = 40$ | $60 - 45 = 15$ | $55 - 40 = 15$ | 40             |

The Minimax Regret choice is Venue C because it has the smallest maximum regret, 20.

### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Venue A        |
| Maximin        | Venue B        |
| Minimax Regret | Venue C        |



## 9.5 Supply Contract

A company must choose among Supply Contract A, Contract B, Contract C, and Contract D. Three states of nature represent different market and supply conditions. The final payoff table, in thousands of dollars, is given below. Use Maximax, Maximin, and Minimax Regret to select a supply contract.

| <i>Payoff table</i> |            |            |            |            |
|---------------------|------------|------------|------------|------------|
| State of nature     | Contract A | Contract B | Contract C | Contract D |
| State $S_1$         | 90         | 80         | 70         | 75         |
| State $S_2$         | 70         | 50         | 45         | 55         |
| State $S_3$         | 60         | 40         | 35         | 50         |

Questions/Tasks.

C4 Applies the Maximin and Maximax criteria

C5 Constructs opportunity loss tables and applies the Minimax regret criterion

#### C4 - Applies the Maximin and Maximax criteria

$$\begin{aligned}\max(\text{Contract A}) &= \max\{90, 70, 60\} = 90, \\ \max(\text{Contract B}) &= \max\{80, 50, 40\} = 80, \\ \max(\text{Contract C}) &= \max\{70, 45, 35\} = 70, \\ \max(\text{Contract D}) &= \max\{75, 55, 50\} = 75.\end{aligned}$$

$$\begin{aligned}\max\{\max(\text{Contract A}), \max(\text{Contract B}), \max(\text{Contract C}), \max(\text{Contract D})\} \\ = \max\{90, 80, 70, 75\} = 90.\end{aligned}$$

The Maximax choice is Contract A because it has the highest possible payoff.

$$\begin{aligned}\min(\text{Contract A}) &= \min\{90, 70, 60\} = 60, \\ \min(\text{Contract B}) &= \min\{80, 50, 40\} = 40, \\ \min(\text{Contract C}) &= \min\{70, 45, 35\} = 35, \\ \min(\text{Contract D}) &= \min\{75, 55, 50\} = 50.\end{aligned}$$

$$\begin{aligned}\max\{\min(\text{Contract A}), \min(\text{Contract B}), \min(\text{Contract C}), \min(\text{Contract D})\} \\ = \max\{60, 40, 35, 50\} = 60.\end{aligned}$$

The Maximin choice is Contract A because it has the largest worst-case payoff.

#### C5 - Constructs opportunity loss tables and applies the Minimax regret criterion

Best payoff in each state:

$$\begin{aligned}\text{State } S_1 &: \max\{90, 80, 70, 75\} = 90, \\ \text{State } S_2 &: \max\{70, 50, 45, 55\} = 70, \\ \text{State } S_3 &: \max\{60, 40, 35, 50\} = 60.\end{aligned}$$

Regret table (best payoff – payoff):

| Alternative | $S_1$          | $S_2$          | $S_3$          | Maximum regret |
|-------------|----------------|----------------|----------------|----------------|
| Contract A  | $90 - 90 = 0$  | $70 - 70 = 0$  | $60 - 60 = 0$  | 0              |
| Contract B  | $90 - 80 = 10$ | $70 - 50 = 20$ | $60 - 40 = 20$ | 20             |
| Contract C  | $90 - 70 = 20$ | $70 - 45 = 25$ | $60 - 35 = 25$ | 25             |
| Contract D  | $90 - 75 = 15$ | $70 - 55 = 15$ | $60 - 50 = 10$ | 15             |

The Minimax Regret choice is Contract A because it has the smallest maximum regret, 0.

#### Decisions summary

| Criterion      | Final decision |
|----------------|----------------|
| Maximax        | Contract A     |
| Maximin        | Contract A     |
| Minimax Regret | Contract A     |

